K L (DEEMED TO BE) UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MARKETPULSE – STOCK RETURN DISTRIBUTION & RISK ANALYTICS SYSTEM

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in Computer Science and Engineering

Submitted by

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Declaration

The Project Report entitled “MARKETPULSE – STOCK RETURN DISTRIBUTION & RISK ANALYTICS SYSTEM” is a record of bonafide work of 2420030753 (Sudhamsha Amaragani) and 2420030776 (Ananya Sinha), submitted in partial fulfillment for the award of B. Tech in CSE to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the project report entitled “MARKETPULSE – STOCK RETURN DISTRIBUTION & RISK ANALYTICS SYSTEM” is a bonafide work done and submitted by 2420030753 (Sudhamsha Amaragani) and 2420030776 (Ananya Sinha) in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of CSE, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Guide

Signature of the Course Coordinator

Signature of the HOD

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ABSTRACT

MarketPulse is a Data Science project that studies how risky a stock has actually been, using nothing more than its own price history. A price chart and an average return, which is what most beginners look at, say very little about this. A share can look calm for months and then give up a large part of its value in a handful of sessions, and the average hides that completely. MarketPulse is built around that gap.

The system takes historical adjusted prices of stocks listed on the National Stock Exchange of India (NIFTY 50 market), either from a local CSV file or downloaded from Yahoo Finance from 1 January 2015, validates them and converts them into daily returns. From the returns it calculates the mean, standard deviation, annual return, annualised volatility, skewness, kurtosis, Historical and Parametric Value at Risk (VaR), Conditional VaR (CVaR), maximum drawdown and the probability of loss, and compares each return distribution with the normal curve. Rolling volatility is used to label market regimes as Calm, Normal, Turbulent or Crisis. K-Means clustering and Principal Component Analysis (PCA) build a Risk DNA profile of each stock, an equally weighted portfolio supports group-level analysis, and historical VaR is evaluated with a 252-observation rolling-window backtest.

Everything is delivered through an interactive Streamlit dashboard built in Python with Pandas, NumPy, SciPy, scikit-learn and Plotly. It has eight tabs (Overview, Distribution Analysis, Risk Metrics, Market Regimes, Risk DNA, Loss Probability, Portfolio Analysis and Risk Report), so a user can pick a stock and read its risk profile without writing any code. In the run documented here, 19 stocks and 2,924 trading days were analysed.

MarketPulse describes the past. It does not forecast prices, it does not recommend what to buy or sell, and it is not an investment-advisory tool. Its estimates rest on historical data from a single market, and its VaR figures are only as reliable as their backtests show. It is an educational and analytical system.

Keywords: Stock return distribution; financial risk analytics; volatility; Value at Risk; Conditional VaR; maximum drawdown; backtesting; K-Means clustering; PCA; market regimes; portfolio analysis; NIFTY 50; Streamlit.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Putting money into shares means accepting that the value can go down as well as up. Most people who look at a stock for the first time check two things: where the price chart has been heading, and what the average return looks like. Both are easy to read, and both are incomplete. The chart shows the path but not how rough it was. The average smooths over every bad week. Two stocks can end a five-year period with exactly the same average return and still have given their holders very different rides.

What separates them is risk, and risk lives in the distribution of returns. How widely do daily returns spread around the average? Are big losses more common than big gains? How often do the extreme days arrive? Questions like these are answered by volatility, skewness, kurtosis, drawdown, Value at Risk and Conditional Value at Risk. Traditional approaches tend to offer these as isolated indicators. Volatility tells us how widely returns vary. Value at Risk gives a sense of the loss we might expect at a chosen confidence level [15]. Conditional Value at Risk goes a step further and describes the average loss beyond the VaR level. Maximum drawdown captures the largest fall from a previous peak. Finance researchers have known for decades that real returns do not follow the neat bell curve of introductory textbooks. Large moves, in both directions, turn up far more often than a normal distribution allows [1], [2]. Anyone relying on a normal-curve picture of risk will therefore be surprised more often than they expect.

The NIFTY 50 is the benchmark index of the National Stock Exchange of India and is made up of fifty large, actively traded companies. Because its constituents are liquid, widely followed and well documented, their price history is a sensible testing ground for a risk-analytics project. MarketPulse was built to bring these views together in one interactive tool. It uses the price history to measure risk statistically, to test whether common risk estimates held up, and to present the outcome in words that a student or a first-time investor can follow.

1.2 Problem Statement

Return data are abundant, but the usual ways of looking at them hide the thing that matters most. A stock that seems steady can lose a large share of its value in a few sessions, and an average return will never show it. Real returns also have heavier tails than the normal curve suggests, so severe losses happen more often than simple models expect.

Measures such as volatility, skewness, kurtosis, drawdown and Value at Risk do exist, but beginners rarely see them explained in everyday language, and they are usually offered one at a time. Many tools either dump raw numbers on the user or attempt to predict future prices, which is unreliable and can mislead. Very few go back and check whether their own risk estimates would have worked on past data.

The problem addressed here is to design and build a single system that (a) cleans historical NIFTY 50 stock data and measures return risk with standard statistics, (b) compares real return distributions with the normal distribution to expose fat-tail behaviour, (c) groups stocks by risk behaviour and separates market periods by condition, (d) estimates Historical and Parametric VaR and CVaR and validates the historical VaR by backtesting, (e) analyses the stocks together as an equally weighted portfolio, and (f) explains the findings through an interactive dashboard, without trying to forecast prices.

1.3 Motivation

- **Looking past the average:** averages are comforting precisely because they hide the bad days. A tool that puts the spread, the skew and the tails in front of the user changes how a stock is judged.
- **Not trusting a number blindly:** a VaR figure printed without any check is just a claim. Backtesting shows how often the estimate was actually broken, which tells the user how far to rely on it.
- **Risk in plain words:** terms like kurtosis put off many beginners. If the dashboard can say the same thing in a sentence anyone can follow, the analysis becomes useful to far more people.

- **Different periods, different behaviour:** a calm year and a crisis year do not behave alike, yet they are often summarised with one set of statistics. Separating the two is more honest.
- **One tool instead of many:** volatility, VaR, CVaR, drawdown, regimes, clustering and portfolio results usually live in separate scripts. Bringing them into one dashboard lets them be compared side by side.
- **A full data-science exercise:** the project covers data collection, cleaning, statistics, unsupervised learning, model validation and a user interface in a single piece of work.

1.4 Scope

MarketPulse analyses the historical behaviour of major Indian stocks and the NIFTY 50 market using daily adjusted price data. Its job is to describe and explain past risk. It does not predict future prices, issue buy or sell signals, or give investment advice, and nothing it produces should be read that way.

The work covers data collection and validation, return calculation, descriptive and distributional statistics, Historical and Parametric VaR, CVaR, maximum drawdown, regime classification, Risk DNA clustering with PCA, holding-period loss probabilities, VaR backtesting, equally weighted portfolio analysis, correlation analysis, an automated risk report and a Streamlit dashboard. Intraday data, derivatives, portfolio optimisation, live trading and the use of news or sentiment are outside the scope of this work and are noted under future enhancements.

1.5 Objectives

1. **Data preparation:** collect and clean historical stock price data (local CSV or Yahoo Finance) for analysis.
2. **Core statistics:** compute daily returns, mean, standard deviation, annual return, annual volatility, skewness, kurtosis and maximum drawdown.
3. **Fat-tail check:** compare return distributions with the normal distribution to spot fat-tail risk.
4. **VaR and CVaR:** calculate Historical and Parametric Value at Risk and Conditional Value at Risk.
5. **Market regimes:** identify market regimes (Calm, Normal, Turbulent, Crisis) using 21-day rolling volatility.
6. **Risk DNA:** classify stocks by historical risk profile using K-Means clustering, and use PCA to visualise stock similarities.
7. **Holding-period loss probability:** estimate the probability of a negative compounded return over horizons from 1 day to 252 days.
8. **Backtesting:** evaluate historical VaR with an out-of-sample rolling-window backtest.
9. **Portfolio analysis:** analyse the risk and performance of an equally weighted portfolio, with correlation analysis.
10. **Risk report and dashboard:** generate a risk report and deliver everything through an interactive Streamlit dashboard.

1.6 Importance and Target Users

The system is meant for students of data science and finance, for beginner investors who want to understand what a stock's history says about risk, and for faculty who want a worked example of statistics applied to real market data. It can also serve as a starting template for a larger risk-analytics tool. It is an educational project. It is not meant to guide real investment decisions on its own, and the dashboard says so openly.

1.7 Organisation of the Report

Table 1.1 – Organisation of the report

Chapter	Focus
1	Introduction – background, problem statement, motivation, scope and objectives
2	Literature Survey – studies on return distributions, regimes, tail risk, clustering and backtesting
3	System and Data Requirements – hardware, software and dataset inventory
4	Technology Stack – libraries and tools used and their roles
5	System Architecture – end-to-end data and module flow
6	Data Understanding and Preparation – dataset, fields, cleaning and returns
7	Statistical Analysis and Feature Engineering – measures, normal comparison and clustering features
8	Risk Modelling Methodology – clustering and PCA, regimes, loss probability, VaR/CVaR, backtesting, portfolio and report generation
9	Implementation and User Interface – project structure, modules, dashboard tabs and screenshots
10	Results and Performance Analysis – observed outputs and their interpretation
11	Testing and Validation – tests and validation status
12	Challenges and Limitations
13	Conclusion and Future Scope
14	References and Appendices

CHAPTER 2 – LITERATURE SURVEY

2.1 Overview

The ideas behind MarketPulse are not new. Each of its modules rests on a body of work that is several decades old, and the project's contribution lies more in bringing them together than in inventing any one of them. The studies reviewed here fall into four groups: work on how returns are distributed, work on markets that switch between different states, work on measuring and validating tail risk, and work on the machine-learning methods used for risk profiling. They shaped four design decisions: to show returns against the normal curve instead of assuming it, to treat periods separately instead of pooling them, to backtest the VaR estimate, and to group stocks by several risk features at once.

2.2 Return Distributions and Stylised Facts

Mandelbrot [1] looked at how speculative price changes are distributed, using cotton prices, and found that large changes happen much more often than a normal distribution would predict. His proposed model was hard to apply in everyday analysis, and later work refined it, but the central observation stuck.

Cont [2] gathered the statistical features that keep appearing in return series across many markets and assets. Heavy tails, volatility clustering (large moves tend to follow large moves) and weak autocorrelation of returns are the main ones. The paper is a description of patterns rather than a ready-made tool, and nothing in it is aimed at a student or small investor. MarketPulse uses these stylised facts as its starting assumptions: it measures skewness and kurtosis for every stock and plots the real distribution against the bell curve.

2.3 Volatility, Value at Risk and Drawdown

Financial risk analysis usually revolves around a handful of measures: volatility, Value at Risk, drawdown and expected shortfall [3], [15], [17]. Volatility is the most common way of showing how much returns vary, and annual volatility makes it easier to compare stocks over a given period. Its weakness is that it treats upward and downward moves alike, so it cannot fully describe downside risk.

Jorion [15] presents Value at Risk as a widely used measure. Historical VaR is taken directly from past returns, whereas Parametric VaR assumes a distribution and uses the mean and standard deviation to set the loss threshold. Glasserman et al. [16] show that heavy-tailed risk factors make VaR harder to estimate, which supports MarketPulse's decision to show Historical and Parametric VaR side by side. Maximum drawdown measures the largest drop in cumulative wealth from a previous peak; unlike volatility, it shows how far the value has actually fallen from its highest historical level.

2.4 Market Regimes, Tail Risk and Validation

Hamilton [3] introduced a regime-switching model in which a series moves between distinct states, each with its own behaviour. The idea carries over to markets: the statistics of a calm period may not describe a crisis at all, so a single set of numbers for the whole history can mislead. Hamilton applied the method to economic output, so it needs adapting for daily stock returns. MarketPulse takes the underlying idea, but uses a simpler rule-based classification of Calm, Normal, Turbulent and Crisis periods from 21-day rolling volatility that a beginner can follow.

Rockafellar and Uryasev [4] studied Conditional Value at Risk, the average loss beyond the VaR level. VaR names a threshold but says nothing about how bad things get once the threshold is crossed. CVaR fills that gap. Their paper concentrates on portfolio optimisation and depends on good tail estimates, which is a reminder that a CVaR figure is only as sound as the data in the tail.

Kupiec [5] proposed a test based on how often actual losses exceed the VaR estimate. If a 95% VaR is breached on much more or much less than 5% of days, the model is suspect. The message is that a VaR model should be validated by backtesting, not simply accepted. MarketPulse follows this idea by comparing the observed breach rate with the expected rate of 5% in a rolling-window backtest; the formal

Kupiec test is kept as a possible extension. Two further references support extensions of the supporting methods: the Jarque–Bera test [6] for checking normality from skewness and kurtosis, and Christoffersen's work on interval forecasts [7], which adds a test of whether VaR breaches cluster together.

2.5 Clustering and Dimensionality Reduction

MarketPulse groups stocks by similar risk profiles. K-Means clustering [18] groups them using standardised return and risk features, and Principal Component Analysis [19] reduces the dimensionality of the feature space so that the groups can be visualised in two dimensions. General treatments of these methods can be found in [20] and [21].

2.6 Summary of Surveyed Studies

Table 2.1 – Surveyed studies and their relevance

Ref.	Authors (year)	Theme	Relevance to MarketPulse
[1]	Mandelbrot (1963)	Distribution of speculative price changes; large moves are more frequent than normal	Motivates the fat-tail comparison
[2]	Cont (2001)	Stylised facts of asset returns: heavy tails, volatility clustering, weak autocorrelation	Basis for skewness, kurtosis and distribution analysis
[3]	Hamilton (1989)	Regime-switching model for non-stationary series	Motivates separating calm and crisis periods
[4]	Rockafellar and Uryasev (2000)	Conditional Value at Risk as the average loss beyond VaR	Basis for the CVaR module
[5]	Kupiec (1995)	Backtest based on frequency of VaR exceptions	Basis for VaR validation by breach rate
[6]	Jarque and Bera (1980)	Normality test using skewness and kurtosis	Possible formal normality check
[7]	Christoffersen (1998)	Evaluation of interval forecasts, including independence of exceptions	Possible extension of the backtest
[15]	Jorion (2006)	Value at Risk as a benchmark for managing financial risk	Basis for Historical and Parametric VaR
[16]	Glasserman et al. (2002)	Portfolio VaR with heavy-tailed risk factors	Supports comparing historical and normal VaR
[17]	Meucci (2005)	Risk and asset allocation	Background for risk and portfolio measures
[18]	MacQueen (1967)	K-means classification of multivariate observations	Basis for Risk DNA clustering
[19]	Jolliffe (2002)	Principal Component Analysis	Basis for the two-dimensional cluster view

2.7 Research Gap and Contribution of MarketPulse

The surveyed work is strong on theory and weak on accessibility. Mandelbrot and Cont describe the patterns but give no tool; Hamilton works with economic output; Rockafellar and Uryasev concentrate on portfolio optimisation; Kupiec supplies a test, not a product. A beginner who wants to look at a real stock through any of these lenses has very little to start with.

MarketPulse contributes in three ways. First, it brings distribution analysis, VaR and CVaR, drawdown, regime classification, Risk DNA clustering, loss-probability heatmaps, backtesting and portfolio analysis together in one interactive dashboard instead of leaving them as separate techniques. Second, it presents the numbers with a plain-language risk report, which the surveyed work does not attempt. Third, it is deliberate about its limits: it explains the past, checks its own estimates against that past and makes no claim to predict the market.

CHAPTER 3 – SYSTEM AND DATA REQUIREMENTS

3.1 Hardware Requirements

MarketPulse is a software project written in Python. Daily data for a few dozen stocks over several years amounts to tens of thousands of rows, which is small by modern standards, and none of the algorithms involved are heavy. K-Means on a few dozen points and rolling backtests on daily series run comfortably on an ordinary laptop, so no GPU or special hardware is needed.

Table 3.1 – Hardware requirements

Resource	Minimum	Recommended
Processor	Dual-core 64-bit CPU	Quad-core 64-bit CPU
Memory (RAM)	4 GB	8 GB or more
Storage	1 GB free (libraries, datasets, saved results)	2 GB free
Display	1366 × 768	1920 × 1080 for comfortable viewing of charts
Network	Not needed to run with stored data	Needed to download data and install packages

3.2 Software Requirements

Table 3.2 – Software requirements

Software	Purpose
Python 3	Implementation language for data handling, statistics, clustering and the dashboard
Pandas, NumPy	Tabular data handling, rolling windows and numerical computation
SciPy	Skewness, kurtosis, normal distribution functions and statistical analysis
scikit-learn	Feature scaling, K-Means clustering and Principal Component Analysis
Plotly	Interactive charts: histograms, box plots, heatmaps, time series and cluster plots
Streamlit	Web framework for the interactive dashboard (served locally during development)
yfinance / Yahoo Finance	Optional; used to download historical prices automatically when no local CSV file is used
Web browser	Viewing the dashboard (any current desktop browser)
Operating system	Windows, macOS or Linux; all support the Python stack above

3.3 Functional Requirements

- The system shall let the user select a stock from the sidebar and view its analysis in separate tabs.
- The system shall calculate and display daily returns, annual return, annual volatility, skewness, kurtosis and maximum drawdown.
- The system shall compare the stock’s return distribution with the normal distribution.
- The system shall calculate Historical VaR, Parametric VaR and CVaR for the stocks.
- The system shall label market periods as Calm, Normal, Turbulent or Crisis using rolling volatility.
- The system shall show which Risk DNA group a stock belongs to, with a PCA view of all stocks.
- The system shall show the historical probability of loss for several holding periods.
- The system shall backtest historical VaR and show the observed and expected breach rates.
- The system shall analyse an equally weighted portfolio and the correlation between stocks.
- The system shall produce a risk report and state clearly that it describes the past and is not investment advice.

3.4 Non-Functional Requirements

- **Usability:** a first-time user should be able to select a stock and read its risk profile in a few steps, with no coding.
- **Reproducibility:** a fixed random state for K-Means and saved data files should make every analysis repeatable.
- **Maintainability:** data loading, metrics, risk analysis, regimes, clustering, reporting and interface live in separate modules.
- **Responsiveness:** results for a single stock should appear quickly enough for interactive use.
- **Transparency:** every figure on the dashboard should be traceable to a documented formula.

3.5 Dataset Inventory

MarketPulse works with publicly available historical adjusted price data of major Indian stocks and the NIFTY 50 market. The data can come from the project dataset (data/stock_prices.csv) or be downloaded from Yahoo Finance by the data loader. The exact download date and the final date range are recorded in Appendix C so that the results can be reproduced.

Table 3.3 – Dataset summary

Detail	Description
Dataset	Historical daily adjusted prices of NSE-listed stocks (NIFTY 50 market)
Source	Local CSV file (data/stock_prices.csv) or Yahoo Finance download
Domain	Indian stock market; financial risk analytics
Default start date	1 January 2015
Records in the documented run	19 stocks analysed over 2,924 trading days (as shown on the Overview tab)
Price used	Adjusted close
Derived fields	Daily return, rolling volatility, drawdown, VaR, CVaR, market regime label, loss probability
Target variable	None. Prices are not predicted; stocks are grouped by clustering
Task	Descriptive statistics, unsupervised grouping, risk estimation and portfolio analysis
End date of the data	[to be recorded at data collection]

3.6 Data Coverage Considerations

Three points about coverage need attention from the start. The index composition changes over time, so a stock in today's NIFTY 50 may not have been in the index years ago, and the analysis covers current constituents (which introduces some survivorship bias, discussed in Chapter 12). Some constituents listed recently and have shorter histories than the rest, so they cannot be treated like long-listed stocks in rolling calculations. Finally, trading calendars differ slightly across data sources, so dates are aligned before any comparison between stocks; the portfolio analysis uses only dates on which every stock has a valid return.

CHAPTER 4 – TECHNOLOGY STACK

MarketPulse is built wholly on open-source Python tools. The table below lists each layer of the stack and what it does in the project.

Table 4.1 – Technology stack

Layer	Technology	Role in MarketPulse
Programming language	Python	Data preparation, statistics, modelling and dashboard logic
Data handling	Pandas, NumPy	Loading price files, computing returns, rolling windows, arrays
Statistics	SciPy	Statistical analysis and the normal distribution functions used for Parametric VaR and the bell-curve overlay
Machine learning	scikit-learn	StandardScaler, K-Means clustering and Principal Component Analysis
Visualisation	Plotly	Interactive histograms, box plots, heatmaps, time series and cluster plots
Dashboard	Streamlit	Interactive web application, sidebar stock selector, tabs and layout
Data storage	CSV files	Saved price data, so the app does not depend on a live feed

4.1 Python

Python is the usual language for data work, and it covers every stage of this project, from reading files to drawing charts to serving a web page. Using one language throughout means no separate front-end technology is needed, and the same functions that the analysis modules use can be called directly by the dashboard.

4.2 Pandas and NumPy

Pandas supplies the DataFrame that holds the price history [9]. Its time-series tools do most of the heavy lifting here: percentage changes for returns, rolling windows for volatility and cumulative maxima for drawdown. NumPy sits underneath and handles the array arithmetic used in quantile estimates.

4.3 SciPy

SciPy provides the statistical functions: sample skewness and kurtosis and the normal distribution’s density and quantile functions (needed to overlay a bell curve and to compute Parametric VaR).

4.4 scikit-learn

scikit-learn offers the StandardScaler used to put the clustering features on a common scale, the K-Means algorithm behind Risk DNA and the Principal Component Analysis used to project the standardised features onto two components. Its consistent fit and transform interface keeps the clustering code short and easy to repeat.

4.5 Plotly

Plotly produces interactive charts that the user can hover over, zoom into and toggle. For a dashboard meant for beginners this matters: being able to point at a bar and see the exact value makes the numbers less intimidating than a static picture would.

4.6 Streamlit

Streamlit turns a Python script into a web application. Selectors, tabs and charts are written in Python without any HTML or JavaScript. During development the application runs locally in a browser.

CHAPTER 5 – SYSTEM ARCHITECTURE

5.1 End-to-End Workflow

The workflow runs in one direction and can be written as P(t) → R(t) → M(t) → A(t), where P(t) stands for historical prices, R(t) for the calculated returns, M(t) for the calculated risk metrics and A(t) for the final analysis outputs. Historical adjusted prices are loaded and cleaned, converted to daily returns and summarised with statistical measures. The returns then feed the risk, regime, clustering, portfolio and report modules side by side. All of them surface through the Streamlit dashboard.

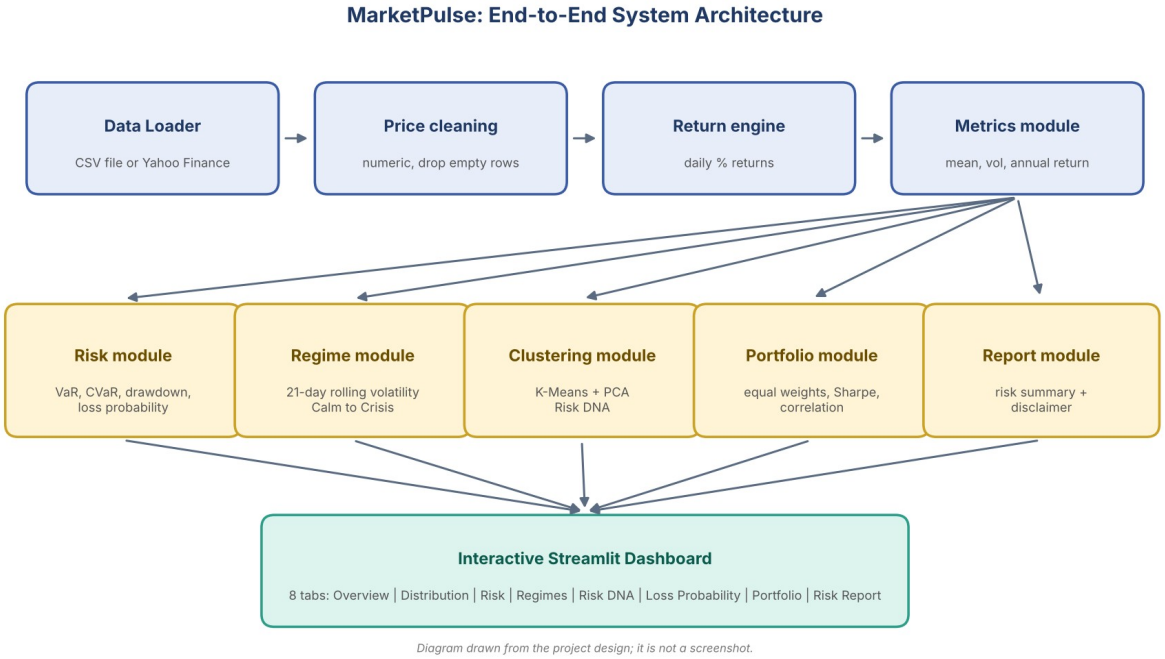


Figure 5.1 – MarketPulse end-to-end system architecture

The cleaned return series feeds the analysis modules, and the dashboard combines their outputs.

5.2 Component Responsibilities

Table 5.1 – Architecture components

Component	Responsibility	Input	Output
Data Loader	Supplies historical adjusted prices from a local CSV file or by download	CSV file / Yahoo Finance	Clean price table
Metrics module	Daily returns and descriptive statistics	Adjusted close prices	Return series, statistics
Risk Analysis module	VaR, CVaR, drawdown, probability of loss, backtest and portfolio metrics	Return series	Risk measures per stock
Regime Detection module	Labels each day as Calm, Normal, Turbulent or Crisis	Rolling volatility	Regime label per day
Clustering module	Scales risk features, runs K-Means and PCA	Risk measures for all stocks	Cluster label, PCA1, PCA2
Portfolio Analysis module	Equally weighted portfolio, Sharpe ratio, correlation matrix	Return series of all stocks	Portfolio metrics
Reporting module	Summarises key risk indicators with a disclaimer	Numerical results	Risk report
Streamlit Dashboard	Collects the stock choice and displays everything	User selections	Charts, tables, report

5.3 Design Principles

- **Explain the past, never predict:** every module describes historical behaviour. No component tries to forecast a price.
- **Check before trusting:** the historical VaR is backtested, and the observed breach rate is compared with the expected one.
- **Plain language alongside numbers:** a figure is shown with some indication of what it means.
- **Modular design:** each analysis metric lives in its own Python module, which makes testing, maintenance and later extension easier.
- **Several views of risk:** volatility, VaR, CVaR, drawdown and loss probability each give a different perspective, and clustering groups stocks by several characteristics at once.

5.4 Data Flow During an Analysis

1. The user opens the dashboard and selects a stock in the sidebar.
2. The price series are loaded from the CSV file or downloaded, converted to numeric values, and rows with only missing values are dropped.
3. Daily returns are computed from adjusted close prices and missing values are removed.
4. The metrics module returns volatility, annual return and descriptive statistics; the risk module returns VaR, CVaR, drawdown and loss probability.
5. The regime, clustering and portfolio results are computed from the same returns.
6. The report module summarises the key indicators, and the dashboard shows charts, tables and text in separate tabs.

CHAPTER 6 – DATA UNDERSTANDING AND PREPARATION

6.1 Historical Stock Dataset

The input is the daily trading history of major Indian stocks and the NIFTY 50 market. Each stock has one row per trading day. The data can be read from the project dataset (data/stock_prices.csv) or downloaded from Yahoo Finance; when a download is needed, the data loader does it automatically. The default start date is 1 January 2015. The data loader also holds the list of stock tickers to be analysed. In the documented run, 19 stocks over 2,924 trading days were analysed. Once the data are saved as a CSV file the analysis does not depend on a live connection and can be repeated exactly.

6.2 Raw and Derived Fields

Table 6.1 – Raw data fields

Field	Description
Date	Trading date
Open / High / Low / Close	Price at the start, highest, lowest and end of the trading day
Adjusted Close	Close price corrected for splits and dividends; used for return calculation
Volume	Number of shares traded

Table 6.2 – Derived fields

Derived field	How it is obtained
Daily return	Percentage change of adjusted close from one day to the next
Annual return	Cumulative growth of the daily returns, annualised with 252 trading days
Rolling volatility	Standard deviation of daily returns over a 21-day moving window, annualised
Drawdown	Percentage fall of cumulative wealth from its running peak
VaR / CVaR	Loss threshold and average tail loss at the 95% confidence level
Market regime label	Calm, Normal, Turbulent or Crisis, based on rolling-volatility percentiles
Holding-period return	Compounded return over 1, 5, 21, 63, 126 or 252 trading days
Risk DNA cluster	Group assigned by K-Means from the stock's standardised risk features

Target variable. MarketPulse has no target to predict. Prices are never forecast. The clustering step is unsupervised, so there is no train–test split of the usual machine-learning kind; instead, historical VaR is validated by a rolling-window backtest (Section 8.5).

6.3 Data Validation and Cleaning

After loading, the data are converted to numeric values and any rows that contain only missing values are dropped. When daily returns are computed, missing values are removed before any further analysis, instead of being filled with invented values, because filling a price would create a fake return. A day on which a stock did not trade is not treated as a zero return.

Stock splits and bonus issues create sudden price drops that are not real losses. Using the adjusted close instead of the raw close avoids counting them as crashes. Extreme returns that remain after adjustment are kept, because the extreme days are exactly what the project is trying to measure. Stocks with a short price history cannot support long rolling windows or long holding periods, so their results should be read with care.

6.4 Return Calculation

Daily returns are computed as simple percentage changes of the adjusted close price using Pandas:

$$R(t) = [\, P(t) - P(t-1) \,] \,/\, P(t-1)$$

where P(t) is the adjusted close on day t and P(t−1) is the adjusted close on the previous trading day. Simple returns are used because they are easy for beginners to read (a return of −3% means the price fell by three percent) and they compound across days into holding-period returns in a straightforward way. These returns feed all of the risk calculations that follow. Annualised figures use 252 trading days per year, a common convention.

CHAPTER 7 – STATISTICAL ANALYSIS AND FEATURE ENGINEERING

This chapter describes the measures calculated for every stock, how return distributions are compared with the normal curve, and the features that are passed to the clustering step. It states what each measure means; the outputs for the documented run appear in Chapters 9 and 10.

7.1 Descriptive Statistics of Returns

Table 7.1 – Statistical measures used

Measure	Definition	What it tells the reader
Mean return	Average of daily returns	Typical daily gain or loss; says nothing about risk
Standard deviation	Spread of daily returns around the mean	How much returns normally vary from day to day
Annual return	Compounded growth of daily returns, annualised	The yearly growth rate implied by the history
Annualised volatility	Daily standard deviation × √252	The same spread expressed on a yearly scale
Skewness	Third standardised moment	Negative: big losses outweigh big gains. Positive: the opposite
Kurtosis (excess)	Fourth standardised moment minus 3	Above zero means fatter tails than a normal curve
Maximum drawdown	Largest fall from a running peak to a later low	The worst peak-to-trough loss a holder would have seen
Probability of loss	Share of days (or holding periods) with a negative return	How often the stock has actually lost money

The annual return and annual volatility are calculated as:

$$R_a = (\prod (1 + R_t))\hat{}(252/n) - 1 \qquad \sigma_a = \sigma_d \times \sqrt{252}$$

where n is the number of daily returns and σ_d is the daily standard deviation. A higher annual volatility means historical returns have varied more. Maximum drawdown is calculated from cumulative wealth W_t = ∏(1 + R_i), the running peak Peak_t = max(W_1, …, W_t) and the drawdown:

$$D_t = W_t / Peak_t - 1 \qquad MDD = min\ over\ t\ of\ D_t$$

7.2 Comparison with the Normal Distribution

The normal distribution is the benchmark. For each stock the real return histogram is drawn with a normal curve that has the same mean and standard deviation on top of it (Figure 9.2). Two things usually show up if the stock behaves like most real assets: a taller, narrower peak, and more observations far out in the tails than the curve allows. Figure 7.1 illustrates the idea with theoretical curves, not real data.

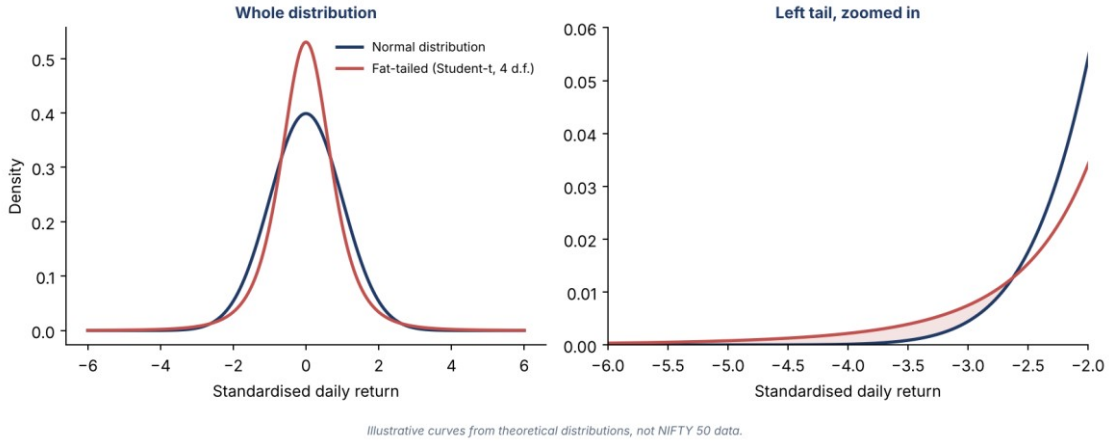


Figure 7.1 – Normal versus fat-tailed return distribution (illustration)

Both curves have the same variance; the fat-tailed curve puts more probability on extreme days. Theoretical illustration only.

Visual comparison is backed by excess kurtosis: a value clearly above zero signals fat tails. A formal Jarque–Bera test [6], which combines skewness and kurtosis into one statistic $JB = (n/6) \times [S^2 + (K - 3)^2/4]$, is a possible extension. Because daily samples are large, such a test would reject normality for almost any real stock, so the size of the departure (the kurtosis itself) matters more than the verdict alone.

7.3 Feature Engineering for Risk DNA

The Risk DNA module groups stocks by how they behave in risk terms, so the features are risk measures, not prices. Raw prices would only separate cheap stocks from expensive ones, which says nothing about risk. For each stock, several financial features are calculated:

Table 7.2 – Features used for Risk DNA clustering

Feature	Why it is included
Annual return	Growth delivered over the period
Annual volatility	Overall level of variability
Historical VaR	Size of a bad day at the 95% level
Historical CVaR	Average loss once the VaR threshold is crossed
Maximum drawdown	Worst historical peak-to-trough loss
Probability of loss	How often daily returns were negative
Average return	Typical daily gain or loss
Skewness and kurtosis	Direction and tail heaviness of the extremes (also shown in the dashboard's cluster table)

Features are standardised before clustering so that a variable measured in large numbers does not dominate the others. Each feature has a plain meaning, which makes the resulting groups easy to describe afterwards.

7.4 Reading the Statistics Carefully

Sample statistics move with the window they are measured on. Skewness and kurtosis are particularly sensitive to a handful of extreme days, so a single crash can change them noticeably. Results therefore need to be read together with the period they cover. A statistic is a description of the sample, not a law of the stock.

CHAPTER 8 – RISK MODELLING METHODOLOGY

The modelling is a chain of well-known statistical and machine-learning methods applied to the return series. This chapter explains each module, the settings used for it and how it is judged. No module forecasts prices.

8.1 Risk DNA: K-Means Clustering and PCA

K-Means [18] divides stocks into k groups so that stocks within a group have similar risk features and stocks in different groups differ. It starts from k initial centres, assigns every stock to the nearest centre, moves each centre to the mean of its stocks and repeats until the assignments stop changing. Its weaknesses are that k has to be chosen and that it prefers roughly round groups.

The standardised features (Table 7.2) are passed to the K-Means algorithm. Up to four clusters can be created, depending on how many stocks are available. The model is initialised with the number of clusters, a random state and multiple initialisations, which keeps the result repeatable. Clusters are ordered by their average volatility and absolute maximum drawdown, and the resulting risk profiles are labelled according to the number of clusters and their relative risk characteristics. The names are only labels for readers; the algorithm itself knows nothing about them.

The standardised features are also passed to Principal Component Analysis [19], and two principal components are kept for visualisation, $X' = XW$, where W is the PCA transformation matrix. PCA1 and PCA2 are then used to visualise how similar the stocks are to one another (Figure 9.5).

8.2 Market Regime Detection

Calm and crisis periods behave very differently, so MarketPulse labels each day with a regime. Inspired by the regime idea in Hamilton's work [3], but kept deliberately simple, the label comes from rolling volatility of the market. If a NIFTY 50 return series is available, it serves as the market return series; otherwise the average return of the selected stocks is used instead. Volatility is computed over a 21-day rolling window, and the resulting volatility distribution is divided into percentile-based categories.

Table 8.1 – Market regime definitions

Regime	Rule (rolling volatility percentile)	Plain meaning
Calm	Below the 25th percentile	Quiet market, small daily moves
Normal	25th to 75th percentile	Typical conditions
Turbulent	75th to 95th percentile	Elevated stress, larger swings
Crisis	Above the 95th percentile	Extreme stress

Days for which the 21-day window is not yet full appear in the dashboard as "Insufficient data". The cut-offs are design choices, not discovered truths, and other regime-detection methods may give different results. A full Markov-switching model is left for future work.

8.3 Holding-Period Loss Probability

Many investors do not hold for one day; they hold for weeks or years. MarketPulse analyses the probability of ending up with a negative compounded return over the horizons 1, 5, 21, 63, 126 and 252 trading days. The compounded return for a horizon h is

$$R(t,h) = \prod (1 + R_i) - 1 \quad (\text{product over } i = t-h+1 \ldots t) \qquad P(\, R(t,h) < 0 \,)$$

and the historical probability of loss is the share of holding periods with a negative return. The results are shown as a bar chart for the selected stock and a heatmap for all stocks (Figure 9.6). It helps users see how the chance of loss changes with the length of the investment horizon, though the data decide whether it shrinks for a given stock. Overlapping windows are not independent, so the percentages describe history and carry no precise statistical confidence.

8.4 Value at Risk and Conditional VaR

Value at Risk at a confidence level is a loss threshold that was not exceeded on that share of days. For a 95% confidence level, α = 0.05, and a one-day VaR of 2.5% means that on 95 out of 100 days the loss was smaller than 2.5%. Conditional VaR is the average loss on the remaining days, those beyond the VaR threshold. Figure 8.1 shows the idea schematically.

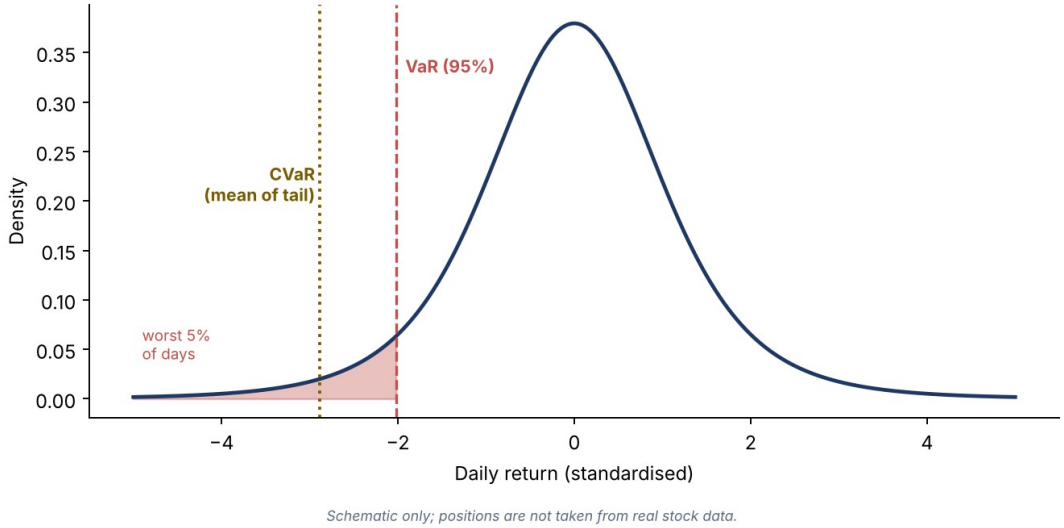


Figure 8.1 – Value at Risk and Conditional VaR (schematic)

VaR marks the edge of the worst tail; CVaR is the average of everything beyond it. Schematic only.

Two methods are used. Historical VaR takes the lower quantile of past returns directly and makes no distributional assumption; Parametric VaR uses the mean μ and standard deviation σ and the inverse standard normal distribution Φ^{-1}:

$$VaR_hist = Q_0.05(R) \qquad VaR_p = \mu + \sigma\, \Phi^{-1}(\alpha) \qquad CVaR_\alpha = E[\, R \mid R \leq VaR_\alpha \,]$$

MarketPulse computes CVaR by averaging the returns that fall below the historical VaR threshold. VaR is a quantile of returns and is therefore a negative number; it is usually quoted as a loss. Putting the two VaR methods next to each other is informative in itself. If the stock has fat tails, the normal method is expected to understate risk, especially at higher confidence levels. Figure 9.3 shows the VaR, CVaR, Parametric VaR and breach-rate table for all stocks.

8.5 Historical VaR Backtesting

MarketPulse runs an out-of-sample historical VaR backtest using a 252-observation rolling window. For every trading day, the VaR threshold is calculated from previous observations only, and the actual return is then compared against it. A VaR breach occurs when R(t) < VaR threshold(t). At a 95% confidence level, the expected breach rate is Be = 0.05 and the observed breach rate is

$$B_o = (number\ of\ VaR\ breaches)\ /\ (number\ of\ test\ observations)$$

Comparing the two shows how well the historical VaR threshold reflects the losses that were actually observed. Too many breaches mean the model understates risk; too few mean it is overly cautious. Breach rates are reported alongside the VaR figures in the Risk Metrics tab (Figure 9.3). Kupiec's proportion-of-failures test [5] and Christoffersen's independence test [7] are possible extensions that would turn this comparison into a formal test.

8.6 Portfolio and Correlation Analysis

MarketPulse analyses an equally weighted portfolio built from the stock return series. Only dates with valid returns for every stock are used. With N stocks, each weight is w_i = 1/N, the daily portfolio return is the weighted sum of the stock returns and the portfolio growth series is its cumulative product:

$$R_{p,t} = \Sigma\, w_i\, R_{i,t} \qquad W_{p,t} = \prod (1 + R_{p,i}) \qquad S = R_{annual} / \sigma_{annual}$$

From these, the portfolio's annual return, annual volatility, Sharpe ratio, historical VaR, historical CVaR and maximum drawdown are calculated. No separate risk-free rate is used in the Sharpe ratio in this implementation. A correlation matrix is also calculated from the stock return data using the Pearson coefficient ρ = Cov(X, Y) / (σ_X σ_Y). A positive correlation means two stocks tend to move in the same direction, while a negative correlation means they tend to move in opposite directions.

8.7 Risk Report Generation

The report module summarises the key risk indicators: annual return, annual volatility, Historical VaR, Historical CVaR, Parametric VaR, maximum drawdown, probability of loss and portfolio risk information. It is rule-based, not a language model, so every statement can be traced back to a visible number. The report states clearly that its values come from historical market behaviour, cannot guarantee future performance and are intended for educational purposes.

8.8 Configuration

Table 8.2 – Configuration settings

Component	Setting	Value
Data	Source; default start date	Local CSV or Yahoo Finance; 1 January 2015
Returns	Type; price used	Simple daily return; adjusted close
Annualisation	Trading days per year	252
Rolling volatility	Window	21 trading days
Regimes	Percentile cut-offs	25 / 75 / 95
Risk DNA	Scaler; algorithm; components	StandardScaler; K-Means; PCA with 2 components
Risk DNA	Number of clusters	Up to 4, depending on the number of stocks
Risk DNA	Initialisations; random state	Multiple initialisations; fixed random state
Holding periods	Lengths (trading days)	1, 5, 21, 63, 126, 252
VaR / CVaR	Confidence level	95% (α = 0.05)
Backtest	Rolling window; expected breach rate	252 observations; 5%
Portfolio	Weights; Sharpe ratio	Equal (1/N); return / volatility, no risk-free rate

8.9 Evaluation Criteria

Table 8.3 – Evaluation criteria for each module

Module	How it is judged
Statistics and normal comparison	Hand-checked against independent calculation; departure from normality read from kurtosis
Risk DNA and PCA	Stability under a fixed random state; whether groups are easy to describe
Regimes	Whether known stressed periods fall into Turbulent or Crisis labels; whether return statistics differ across regimes
Loss probability	Agreement with a direct count on sample cases
VaR / CVaR	Observed breach rate compared with the expected 5% rate
Portfolio	Equal weights sum to one; results agree with an independent calculation
Risk report	Each statement traceable to a number; no unsupported statements
Dashboard	Correct display of all outputs; tasks completed by test users without help

CHAPTER 9 – IMPLEMENTATION AND USER INTERFACE

9.1 Implementation Overview

MarketPulse is developed in Python. The application is divided into several modules to keep it organised, easy to maintain and easy to extend. The project structure is:

```
MarketPulse/
|
|-- app.py
|
|-- data/
|   |-- stock_prices.csv
|
|-- src/
|   |-- __init__.py
|   |-- data_loader.py
|   |-- metrics.py
|   |-- risk.py
|   |-- regimes.py
|   |-- clustering.py
|   |-- report.py
|
|-- requirements.txt
|-- implementation_screenshots/
|   |-- 01_market_overview.png
|   |-- 02_distribution_analysis.png
|   |-- 03_risk_metrics.png
|   |-- 04_market_regimes.png
|   |-- 05_risk_dna_clustering.png
|   |-- 06_loss_probability.png
|-- MarketPulse_IEEE_Report.tex
```

9.2 Software Modules

Table 9.1 – MarketPulse software modules

Module	Function
app.py	Main application and Streamlit interface
data_loader.py	Loads or downloads historical market data
metrics.py	Calculates return and statistical metrics
risk.py	Calculates risk, VaR, CVaR, drawdown, loss probability and portfolio metrics
regimes.py	Detects market regimes using rolling volatility
clustering.py	Performs K-Means clustering and PCA analysis
report.py	Generates risk analysis reports

9.3 Dashboard Tabs

MarketPulse offers an interactive web dashboard built with Streamlit. A sidebar lets the user pick a specific stock, and the analysis is organised into eight tabs:

Table 9.2 – Dashboard tabs and elements

Tab	Purpose
1. Overview	Number of stocks analysed, number of trading days, selected stock and an annual volatility comparison drawn as an interactive Plotly chart, with a statistical summary

Tab	Purpose
2. Distribution Analysis	Mean daily return, daily volatility and observations; return histogram with normal curve; box plot
3. Risk Metrics	Value at Risk and Conditional VaR table (Historical VaR, Historical CVaR, Parametric VaR, breach rates, max drawdown) and maximum drawdown comparison
4. Market Regimes	Market volatility and regimes over time, with the regime distribution (number of days per regime)
5. Risk DNA	K-Means clusters visualised using PCA, and a cluster characteristics table
6. Loss Probability	Holding-period loss probability for the selected stock and an all-stock heatmap
7. Portfolio Analysis	Equally weighted portfolio metrics and correlation analysis
8. Risk Report	Summary of key risk indicators with the educational disclaimer

9.4 Implementation Evidence

The figures that follow (Figures 9.1 to 9.6) were captured from the running Streamlit application with RELIANCE.NS selected. They cover the Overview, Distribution Analysis, Risk Metrics, Market Regimes, Risk DNA and Loss Probability tabs.

9.4.1 Overview

The Overview tab (Figure 9.1) reports 19 stocks analysed over 2,924 trading days, with RELIANCE.NS as the selected stock. Below the headline figures, an interactive bar chart compares the annualised volatility of all stocks, followed by a statistical summary table.

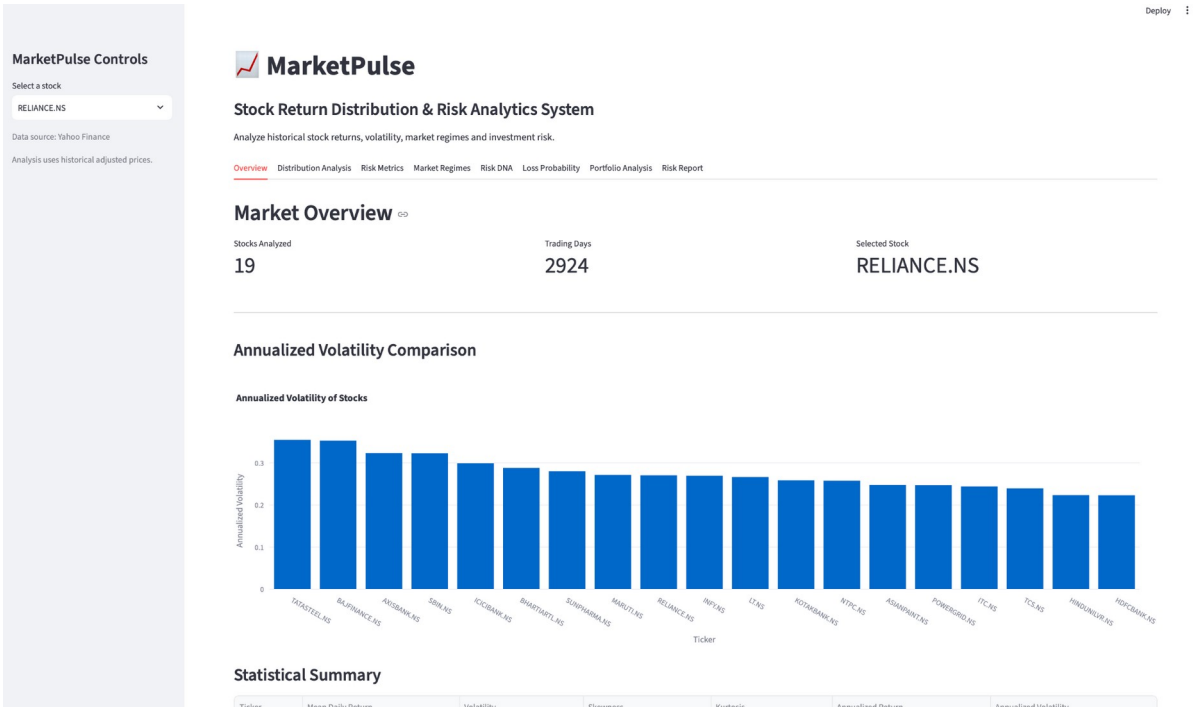


Figure 9.1 – Overview tab: stocks analysed, trading days and annualised volatility comparison

9.4.2 Distribution Analysis

The Distribution Analysis tab (Figure 9.2) shows the stock return distribution for the selected stock: a mean daily return of 0.0808%, a daily volatility of 1.7041% and 2,924 observations, together with the return histogram and fitted normal curve and a box plot.

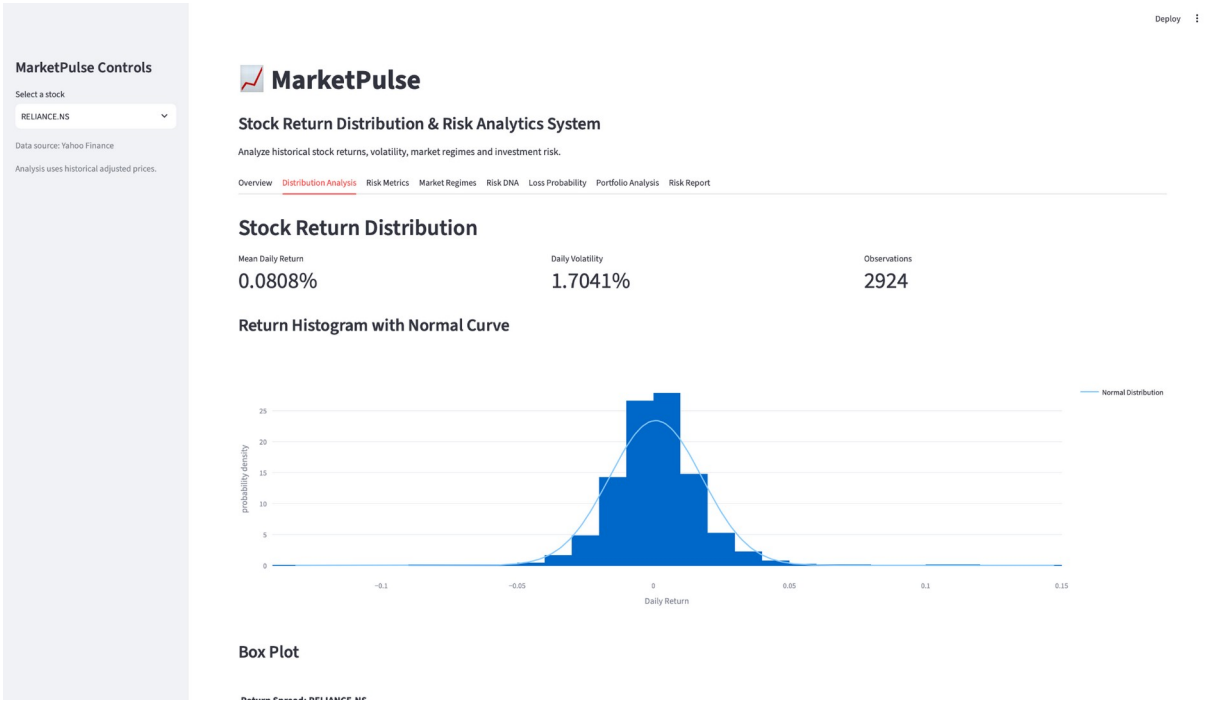


Figure 9.2 – Distribution Analysis tab: return histogram with fitted normal curve

9.4.3 Risk Metrics

The Risk Metrics tab (Figure 9.3) lists Value at Risk and Conditional VaR for every stock, followed by a maximum drawdown comparison. VaR gives a loss threshold, CVaR gives the average loss beyond it, and maximum drawdown offers another view of historical downside behaviour.

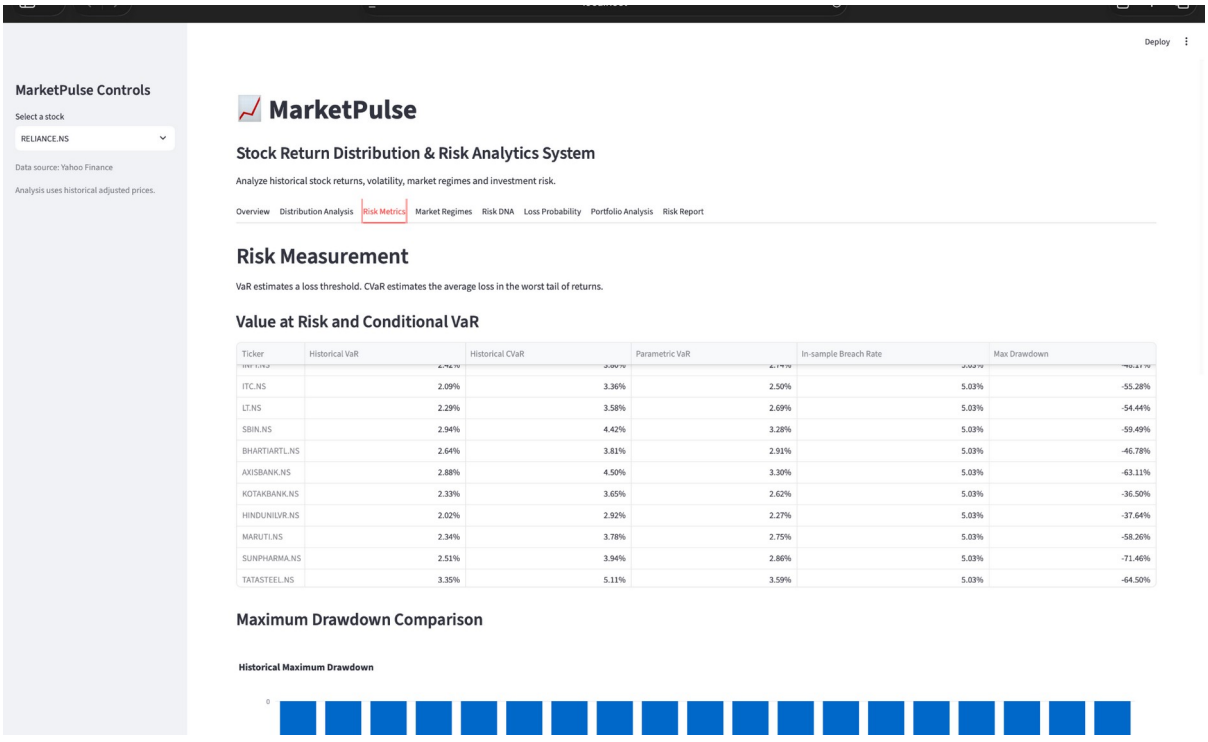


Figure 9.3 – Risk Metrics tab: Value at Risk, CVaR and maximum drawdown

9.4.4 Market Regimes

The Market Regimes tab (Figure 9.4) classifies market conditions using 21-day rolling annualised volatility, plots volatility coloured by regime and tabulates the number of days in each regime.

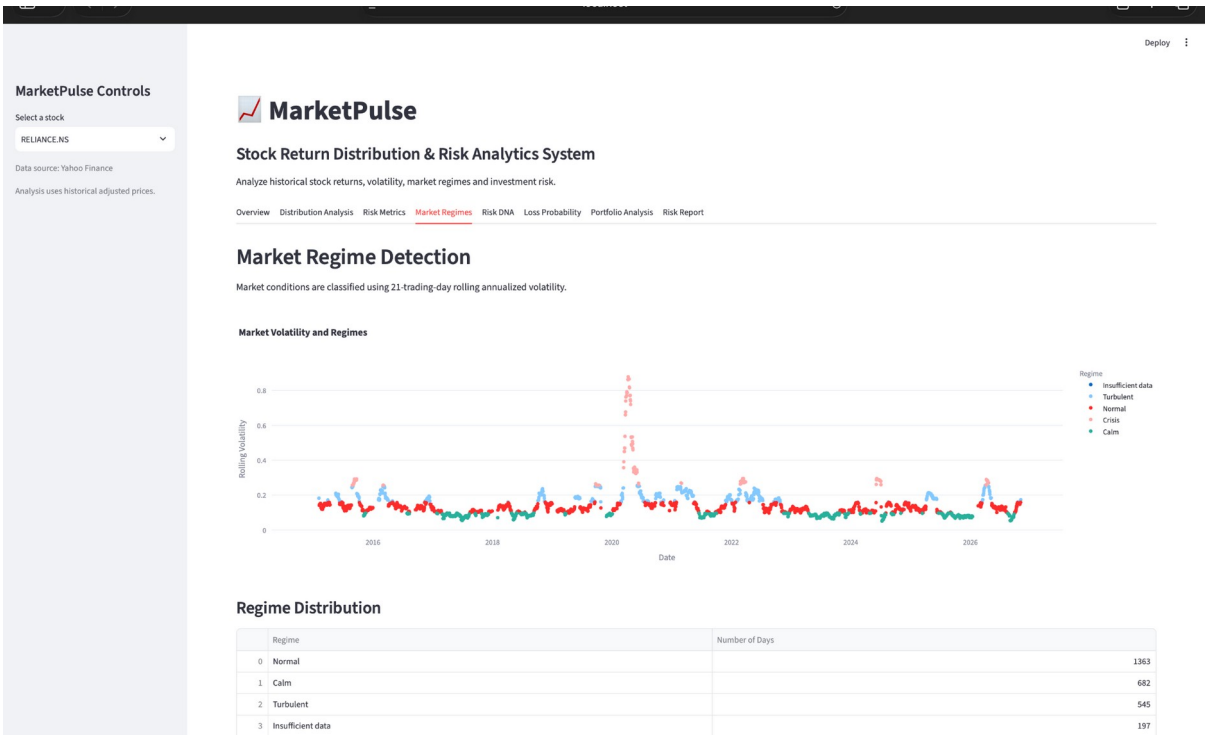


Figure 9.4 – Market Regimes tab: rolling volatility and regime detection

9.4.5 Risk DNA

The Risk DNA tab (Figure 9.5) groups stocks by their historical return and risk characteristics using K-Means. The PCA plot shows how similar the stocks are, and the cluster characteristics table gives each stock's features, cluster and risk profile.

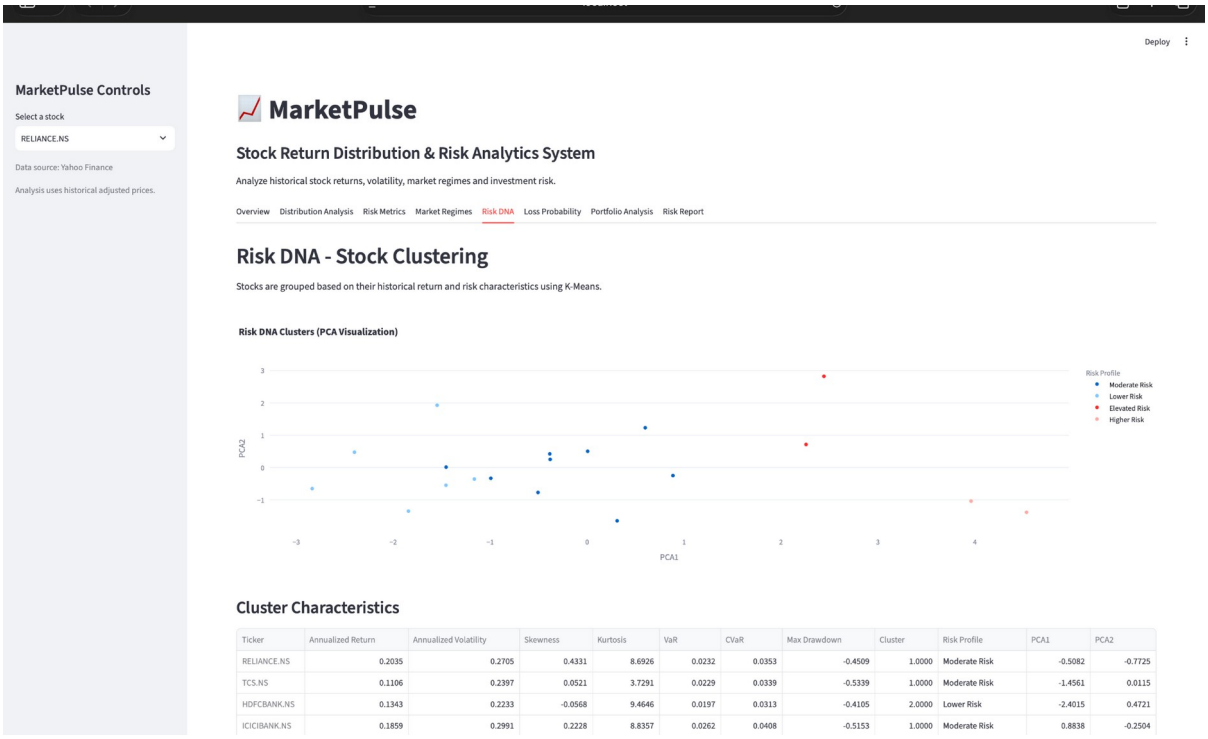


Figure 9.5 – Risk DNA tab: K-Means risk clusters visualised using PCA

9.4.6 Loss Probability

The Loss Probability tab (Figure 9.6) shows how frequently historical holding-period returns were negative: a bar chart of loss probability by holding period for RELIANCE.NS, and a heatmap across all stocks and holding periods.

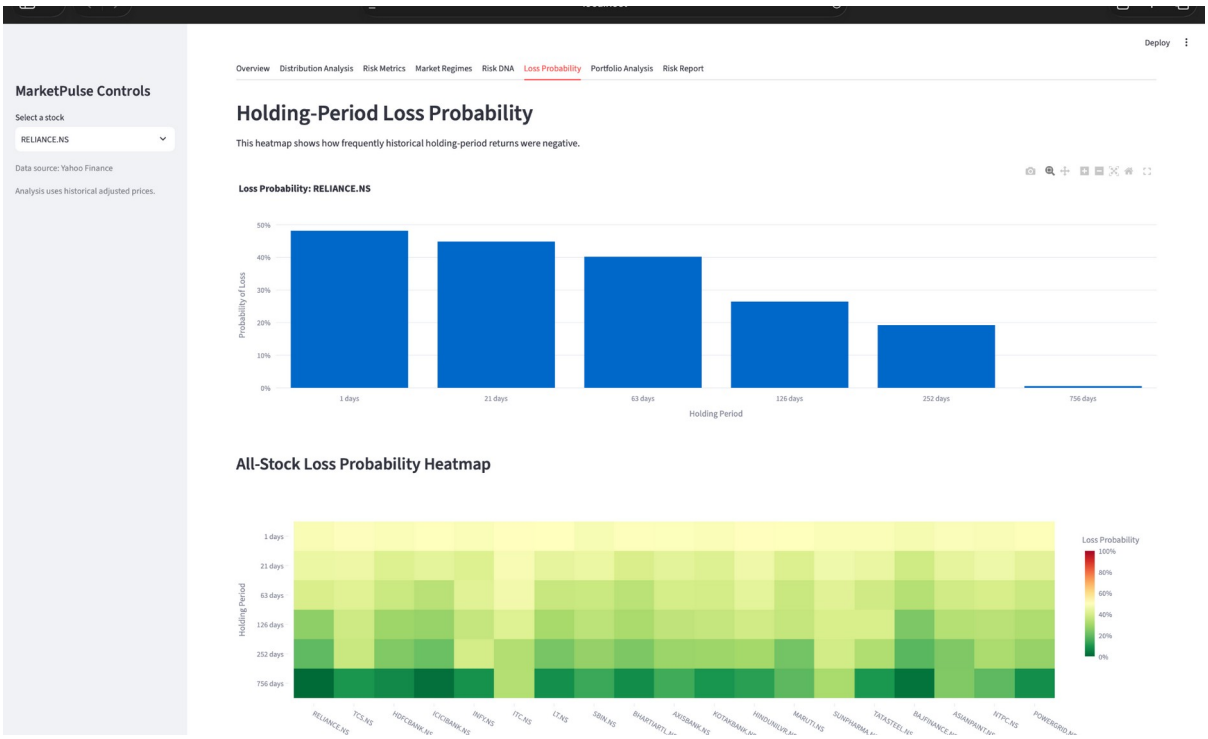


Figure 9.6 – Loss Probability tab: holding-period loss probability and heatmap

The Portfolio Analysis and Risk Report tabs are part of the application; screenshots of these two tabs may be added from the running dashboard if the examiners require them.

9.5 User Journey

1. Open MarketPulse and read the introduction and disclaimer.
2. Select a stock in the sidebar.
3. Look at the Overview tab and the annual volatility comparison.
4. Open the Distribution Analysis tab and compare returns with the normal curve.
5. Check VaR, CVaR and drawdown in the Risk Metrics tab, and the regime history in Market Regimes.
6. Check the stock's Risk DNA group, the loss-probability heatmap and the portfolio results.
7. Finish with the Risk Report, keeping in mind that it describes the past only.

CHAPTER 10 – RESULTS AND PERFORMANCE ANALYSIS

MarketPulse produces a variety of analysis outputs from historical stock return data. This chapter summarises what each output shows and how it should be read. The dashboard screenshots are given in Chapter 9 (Figures 9.1 to 9.6).

10.1 Summary of Outputs

Table 10.1 – Summary of dashboard outputs

Output	Observed in the documented run	Figure
Dataset size	19 stocks analysed over 2,924 trading days	9.1
Return distribution	RELIANCE.NS: mean daily return 0.0808%, daily volatility 1.7041%, 2,924 observations	9.2
VaR, CVaR, drawdown	Table of Historical VaR, Historical CVaR, Parametric VaR, breach rates and maximum drawdown for all stocks	9.3
Market regimes	Rolling-volatility timeline coloured by Calm, Normal, Turbulent, Crisis, with days per regime	9.4
Risk DNA	K-Means clusters in PCA space, with a cluster characteristics table	9.5
Loss probability	Bar chart for RELIANCE.NS and heatmap for all stocks over six holding periods	9.6

10.2 Volatility Comparison

The annual volatility comparison (Figure 9.1) helps pick out stocks with relatively high or low historical volatility. Stocks with higher volatility have shown larger swings in their daily returns. Still, volatility should not be treated as a complete measure of risk, since it makes no distinction between positive and negative price movements.

10.3 Return Distribution

The return distribution (Figure 9.2) shows how often returns of different sizes have occurred. It can help reveal skewness, extreme values and differences in return behaviour between individual stocks [22]. Comparing the histogram with the fitted normal curve makes any fat-tail behaviour visible.

10.4 Risk Metrics

The risk analysis (Figure 9.3) draws on Historical VaR, Historical CVaR, Parametric VaR and maximum drawdown. VaR gives a loss threshold, CVaR gives the average loss beyond that threshold, and maximum drawdown offers another view of historical downside behaviour. Placing Historical and Parametric VaR side by side shows how far the normal assumption departs from what was actually observed.

10.5 Market Regimes

Rolling volatility (Figure 9.4) is an intuitive way to spot changes in market conditions. The four regimes make it easier to understand historical periods with different levels of volatility.

10.6 Risk Profiling

K-Means clustering groups stocks with similar historical return and risk characteristics, and PCA provides a two-dimensional view of the standardised risk features (Figure 9.5). Because only a limited number of stocks are available, group boundaries are not sharp.

10.7 Holding-Period Risk

Loss probability (Figure 9.6) shows how often negative compounded returns have occurred over different investment horizons. This gives a deeper look at how losses have behaved across different holding periods.

10.8 Backtesting and Portfolio Results

The historical VaR backtest compares the observed breach rate with the expected rate of 5% (Section 8.5), and the portfolio module reports the annual return, annual volatility, Sharpe ratio, VaR, CVaR and maximum drawdown of the equally weighted portfolio (Section 8.6). These outputs are produced by the Risk Metrics and Portfolio Analysis tabs.

10.9 What the Results Should Be Read Against

Literature gives some expectations to test. Studies of asset returns report positive excess kurtosis in most series [2], so most stocks should show fat tails. If so, the normal-method VaR is expected to understate risk compared with the historical method. Stressed regimes should show higher volatility and wider VaR than calm ones by construction, so the more interesting question is whether the shape of the distribution also changes. The loss heatmap should show whether longer holding periods have reduced the historical chance of loss, though a bull-market sample could make this look better than it would be in general.

All results carry the same reminder: they describe the chosen stocks over the chosen period and say nothing certain about the future.

CHAPTER 11 – TESTING AND VALIDATION

11.1 Testing Approach

Two kinds of evidence are collected. The first is correctness of the calculations: each formula is checked on small series where the answer is known in advance, and on real stocks against an independent tool. The second is functional testing of the dashboard: walking through the user journey and checking each tab against the underlying numbers. In addition, historical VaR is validated statistically through the rolling-window backtest described in Chapter 8.

11.2 Verification Tests

Table 11.1 – Verification tests

Test ID	Area	Test description	Expected result	Status
TC-01	Data loading	Load a file with a missing price	Rows with only missing values are dropped	Planned
TC-02	Returns	Compute returns for a short hand-made price series	Matches hand calculation	Planned
TC-03	Drawdown	Apply drawdown to a series with a known peak and trough	Reported value equals hand value	Planned
TC-04	Skewness / kurtosis	Compare with an independent tool on one stock	Values agree to rounding	Planned
TC-05	Normal comparison	Run on data simulated from a normal distribution	Excess kurtosis near zero	Planned
TC-06	Risk DNA	Repeat clustering with the same random state	Identical groups every run	Planned
TC-07	Regimes	Check labels around a known stressed period	Period labelled Turbulent or Crisis	Planned
TC-08	Loss probability	Count losing windows by hand for a short series	Probability matches the module	Planned
TC-09	VaR / CVaR	Compute historical VaR on a small sample by hand	Matches module output	Planned
TC-10	Backtest	Run the backtest on simulated data with a known exception rate	Observed breach rate close to the expected rate	Planned
TC-11	Portfolio	Check that equal weights sum to one and compare with a manual average	Portfolio return matches manual calculation	Planned
TC-12	Dashboard run	Run the Streamlit app with RELIANCE.NS selected and open each tab	Overview, Distribution, Risk, Regimes, Risk DNA and Loss Probability tabs render (Figs. 9.1–9.6)	Completed

11.3 System-Level Tests

Table 11.2 – System-level tests

Test area	Description	Status
Dashboard flow	Select several different stocks and check that every tab updates correctly	Planned
Edge cases	Very short date range, stock with short history, period with no regime change	Planned
Cross-browser and device checks	Rendering on several browsers and screen sizes	Planned
Performance	Response time when switching stocks	Planned
Usability	Short trial with classmates unfamiliar with the project	Planned

Test area	Description	Status
Out-of-sample check	Run the pipeline on a different time window or a second market index	Planned

11.4 Validation Status

The dashboard has been run end to end and its tabs captured as screenshots (Chapter 9). Formal test results are not yet recorded in this report, and the status columns above are to be updated as tests are carried out. The system is validated on historical data from Indian stocks only, so conclusions about other markets, asset classes or future periods are not established and should not be assumed.

CHAPTER 12 – CHALLENGES AND LIMITATIONS

12.1 Challenges Encountered

- **Data quality:** public price data can contain gaps, adjusted values that differ between sources and the odd obviously wrong entry. Each needs a decision about whether to fix, drop or flag it.
- **Heavy-tailed data:** common summary statistics assume well-behaved data. For fat-tailed returns they become unstable, which complicates both estimation and interpretation.
- **Choosing settings without overfitting:** the number of clusters, window lengths and regime cut-offs all involve judgement, and it is tempting to tune them until the results look good.
- **Short samples in the tail:** a VaR estimate rests on few observations in the tail, so it is noisy.
- **Explaining numbers simply:** turning kurtosis or CVaR into a sentence that is both accurate and plain is harder than it looks.
- **Joining separate modules:** keeping the data, the computed results and the dashboard consistent with each other takes careful design.

12.2 Limitations

1. **Past is not future:** the analysis is based on historical market data and cannot guarantee future market behaviour.
2. **Parametric VaR assumes normality:** a normal distribution may not represent financial returns well, especially in extreme market conditions.
3. **Equally weighted portfolio:** this is a simple construction method and does not represent all possible investment strategies. The Sharpe ratio uses no risk-free rate.
4. **Rule-based regimes:** market regime detection relies on rolling-volatility percentiles chosen by the team, and other regime detection methods may give different results.
5. **Clustering assumptions:** clustering depends on the chosen features, the preprocessing method and the number of clusters; with few stocks, group boundaries are not sharp.
6. **Single market:** only Indian stocks are analysed, so the findings may not transfer to other markets or to smaller companies.
7. **Survivorship bias:** using today's index constituents leaves out companies that were removed or delisted, which makes historical risk look slightly smaller than it was.
8. **Heatmap overlap:** holding-period windows overlap heavily, so the loss probabilities are descriptive percentages without strong statistical confidence.
9. **No transaction costs or taxes:** none of the figures take trading costs, taxes or liquidity into account.
10. **Educational prototype:** the application is intended primarily for educational purposes, has not been reviewed by finance professionals and has not been deployed for public use.

12.3 Ethical and Safety Considerations

Financial tools can do harm if people take them for more than they are. MarketPulse is presented plainly as an analytical and educational system. It does not predict prices, it does not tell anyone to buy or sell, and the risk report states that it describes the past. The language of the generated text is kept careful: it states what happened, not what will happen. Any later version meant for real users would need clear disclosure of uncertainty, review by qualified professionals and a check against the regulations that apply to investment-related information.

CHAPTER 13 – CONCLUSION AND FUTURE SCOPE

13.1 Conclusion

MarketPulse started from a simple observation: the way most people look at stocks tells them very little about risk. The project responds with a complete system for stock return distribution and risk analysis. It calculates daily returns from historical price data and applies a range of financial analytics techniques to study how stocks behave: annual return, annual volatility, Historical VaR, Conditional VaR, Parametric VaR, maximum drawdown, market regime detection, risk profiling with K-Means and PCA, holding-period loss probability, VaR backtesting and portfolio analysis.

The Streamlit dashboard lets users explore the results interactively through separate tabs, and the modular Python implementation keeps the system easy to maintain and extend. What sets the design apart is less any single technique than the combination, and the discipline around it: nothing is predicted, the VaR estimate is checked against a backtest, and the risk report states that it describes the past. Overall, MarketPulse shows how historical stock data can be combined with statistical analysis, risk measurement, clustering, dimensionality reduction and interactive visualisation to give a comprehensive picture of historical market risk.

13.2 Status Summary

Table 13.1 – Implementation status

Component	Status	Description
Problem definition and objectives	Completed	Problem statement, objectives, scope and gap identification
Literature survey	Completed	Core studies reviewed and related to each module
System design and workflow	Completed	Architecture, modules and data flow defined
Data collection and cleaning	Completed	Data loader reads the CSV file or downloads from Yahoo Finance; returns computed
Statistical analysis and normal comparison	Completed	Descriptive statistics, annual return and volatility, histogram with normal curve
Risk DNA, regimes, loss probability, VaR/CVaR	Completed	K-Means and PCA, regime detection, loss heatmap, Historical and Parametric VaR, CVaR, backtest
Portfolio analysis and risk report	Completed	Equally weighted portfolio, correlation, Sharpe ratio and report module
Dashboard	Completed	Streamlit application with eight tabs
Integration and testing	In progress	Dashboard run verified (Chapter 9); formal test results to be recorded
Finalisation and documentation	In progress	Project report and IEEE-format research paper prepared

13.3 Future Enhancements

- Add more market and financial datasets, including mutual fund analysis and sector-level risk analysis.
- Add alternative portfolio weighting strategies and more portfolio optimisation techniques.
- Replace the rule-based regimes with a Markov regime-switching or hidden Markov model and compare the two, or add other advanced regime detection methods.
- Add formal backtests: the Kupiec test for the exception rate, the Christoffersen test for clustering of exceptions and a Basel-style traffic-light summary.
- Include other VaR methods such as Cornish–Fisher or volatility-weighted estimation.
- Add more machine learning models and add rolling re-clustering so that a stock's Risk DNA group can be followed over time.

- Re-run the pipeline on other indices and markets, and on a point-in-time constituent list to reduce survivorship bias.
- Improve the automated risk reports and provide more visual analytics.
- Support real-time or near-real-time market data and automatic refresh, and deploy the dashboard with a clear disclaimer and usage logging.
- Test the readability of the reports with real beginners and improve the wording accordingly.

13.4 Skills Demonstrated

The project involves work with financial time-series data, data cleaning, descriptive and inferential statistics, unsupervised machine learning, model validation by backtesting, portfolio analysis, interactive visualisation and Streamlit development, along with attention to how risk information should be communicated responsibly.

CHAPTER 14 – REFERENCES AND APPENDICES

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14.2 Appendices

Appendix A describes the data fields and derived measures, Appendix B lists the abbreviations used, Appendix C records the settings needed to reproduce the analysis, and Appendix D gives the location of the source code.

Appendix A – Field and Feature Descriptions

A.1 Raw fields

Field	Description
Date	Trading date
Open	Opening price
High	Highest price of the day
Low	Lowest price of the day
Close	Closing price
Adjusted Close	Closing price adjusted for splits and dividends
Volume	Shares traded

A.2 Derived measures

Measure	Description
Daily return	Percentage change of adjusted close between consecutive days
Mean return	Average daily return over the period
Standard deviation	Spread of daily returns
Annual return	Compounded growth of daily returns annualised with 252 trading days
Annualised volatility	Daily standard deviation multiplied by the square root of 252
Skewness	Asymmetry of the return distribution
Excess kurtosis	Tail heaviness relative to the normal distribution
Maximum drawdown	Largest percentage fall from a running peak
Historical VaR	Lower 5% quantile of historical returns
Parametric VaR	Loss threshold from the mean, standard deviation and the normal quantile
CVaR	Average loss on days beyond the VaR level
Probability of loss	Share of holding periods with a negative compounded return
Sharpe ratio	Annual return divided by annual volatility (no risk-free rate)
Regime label	Calm, Normal, Turbulent or Crisis
Cluster (Risk DNA)	Risk group assigned by K-Means
PCA1, PCA2	First two principal components of the standardised features

Appendix B – Abbreviations

Abbreviation	Meaning
CSV	Comma-separated values
CVaR	Conditional Value at Risk
JB	Jarque–Bera (normality test)
MDD	Maximum drawdown
NIFTY 50	Benchmark index of 50 large companies on the National Stock Exchange of India
NSE	National Stock Exchange of India
OHLCV	Open, High, Low, Close, Volume
PCA	Principal Component Analysis
POF	Proportion of failures (Kupiec test)
VaR	Value at Risk

Appendix C – Reproducibility Settings

Setting	Value
Data source	Local CSV (data/stock_prices.csv) or Yahoo Finance download; download date [to be recorded]
Date range analysed	From 1 January 2015; 2,924 trading days; end date [to be recorded]
Stocks included / excluded	19 stocks analysed in the documented run [list to be recorded, with reasons for any exclusions]
Return definition	Simple daily return from adjusted close
Trading days per year	252
Rolling volatility window	21 trading days
Risk DNA clusters (k)	Up to 4, depending on the number of stocks; fixed random state; multiple initialisations
Regime cut-offs	25 / 75 / 95 percentiles of rolling volatility
VaR confidence level and window	95%; 252-observation rolling window for the backtest
Holding periods	1, 5, 21, 63, 126, 252 trading days
Library versions	[Python, Pandas, NumPy, SciPy, scikit-learn, Plotly, Streamlit – to be recorded]

Appendix D – Source Code Repository

GitHub repository: [add repository link]